

XAI in the Context of Predictive Process Monitoring: An Empirical Analysis Framework

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Research Paper Presentation / CPSC 5383 Course Work

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Agenda

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Problem Statement

The Black Box Challenge in Predictive Process Monitoring (PPM)

Problem: ML models in PPM lack transparency, making predictions hard to trust.

Core Research: Analyzing XAI methods to ensure consistent, robust, and actionable explanations in PPM.

Inputs: Event logs, preprocessing techniques, ML models, XAI methods.

Process: Compare XAI methods under varying PPM settings.

Output: Insights into explanation stability, robustness, and efficiency.

Why is Explainability Important?

Trust and Adoption: Stakeholders need to trust predictions, especially in high-stakes domains like healthcare.

Debugging and Improvement: Explanations reveal data issues or model biases, enabling process improvement.

Regulatory Compliance: Explainability is required by laws like GDPR.

User Engagement: Transparent predictions enhance user engagement and advocacy.

Innovation: XAI Framework in PPM

A Unified framework to analyze XAI methods under varying PPM settings.

Inputs: Event logs (e.g., Sepsis, Traffic Fines), preprocessing (e.g., index-based encoding), models (e.g., XGBoost, Logistic Regression), XAI methods (e.g., SHAP, LIME).

Process: Evaluate global and local explanations for consistency, robustness, and efficiency.

Outputs: Comparative analysis of XAI methods, insights into preprocessing impact, and recommendations for best practices.



Data Dimension (before preprocessing)

- Data characteristics analysis (Correlation- Mutual Information - Profiling)

Preprocessing Dimension

- Single (Bucketing)+ Aggregation (Encoding)
- Prefix (Bucketing) + Index (Encoding)

ML Model Dimension

- XGBoost
- Logistic Regression (LR)

XAI Dimension (Global)

- XAI methods (Permutation Importance - SHAP - ALE)
- Compare execution times
- Check stability

XAI Dimension (Local)

- XAI methods (LIME - SHAP)

Benefits & Impact

Real-World Impact of Explainable PPM

Benefits:

- Consistent and robust explanations.
- Identification of data issues (e.g., collinearity).
- Faster execution times for scalable solutions.

Use Cases:

- Healthcare: Explaining sepsis predictions to doctors.
- Finance: Justifying loan approval processes to auditors.
- Government: Clarifying traffic fine predictions to citizens.

Where Do We Go from Here?

Gaps:

Limited XAI methods analyzed.

Small dataset sizes in experiments.

Lack of user studies to assess explanation utility.

Further Research Questions:

Q1: How can explanation stability be mathematically formalized?

Q2: What's the impact of large datasets on explanation quality?

Q3: Can we develop a unified metric for explanation utility?

Making Explanations Consistent

Research Question (Q1): How can explanation stability be mathematically formalized and optimized?

Mathematical Formulation:

$$\min_{\theta} \mathbb{E}_{x \sim D} [\text{Var}(\phi(f, (x; \theta)x))]$$

Algorithm: StableXAI – Generates stable explanations by minimizing variance.

Inputs: Dataset D , Model f , XAI method ϕ , Number of runs N .

Output: Stable explanation ϕ_{stable} .

StableXAI Algorithm

Algorithm: StableXAI

Input: Dataset D , Model f , XAI method ϕ , Number of runs N

Output: Stable explanation ϕ_{stable}

Initialize: Explanation list $E = []$

For $i = 1$ to N :

 Generate explanation $\phi_i = \phi(f, D)$

 Append ϕ_i to E

End For

Compute variance $\text{Var}(E)$ for each feature in explanations

Select features with $\text{Var}(E) < \text{threshold}$ as stable features

Construct ϕ_{stable} using stable features

Return ϕ_{stable}

Evaluation

Stability Metric:
$$\text{Stability Score} = 1 - \frac{\text{Var}(\phi(f(x; \theta)x))}{\text{Max Var}}$$

Where *Max Var* is the maximum possible variance.

Consistency Check: Compare explanations across multiple runs using similarity metrics (e.g., Jaccard similarity for feature sets).

Computational Efficiency: Measure execution time and scalability with dataset size.

User Study: Conduct surveys or interviews with stakeholders to assess the utility and trustworthiness of stable explanations.

Robustness Test: Evaluate stability under varying conditions (e.g., different preprocessing techniques, imbalanced datasets).



Thank You!